**34** When a thin gold foil is bombarded with alpha particles, some alpha particles are deflected through large angles.

Which statement explains this deflection?

**A** Most of the atom consists of empty space.

**B** All of the positive charge and most of the mass of the gold atom are concentrated in a small volume.

**C** Positive charge in the gold atom is spread evenly throughout the atom.

**D** All of the negative charge is concentrated at its centre.